AAROHAN 2026
ASCENT OF KNOWLEDGE

Class 9th

Name

Contact Number

BITS Pilani
Pilani Campus

APOGEE 2026

Instructions:

1. This Question Paper contains 4 comprehensive passages based on the above theme. Each passage is followed by a series of questions. There will be 25 questions in all.

2. Each question has **ONLY ONE** out of the four choices as the correct answer. Marking Scheme is
    a. **+4 for correct Answer**
    b. **-1 for incorrect Answer**
    c. **and 0 for not attempting the question.**.

3. The questions will test your skills in Physics, Chemistry, Mathematics and Analytical Reasoning, along with your understanding of the passage. Remember, everything you need to know is already given somewhere in this passage- do not be intimidated, for winners in this game are just scared stupid explorers.

ALL THE BEST (ノ ﾟｰﾟ)ノ

Space for Rough Work

RESOURCES LOW, MORALE LOWER

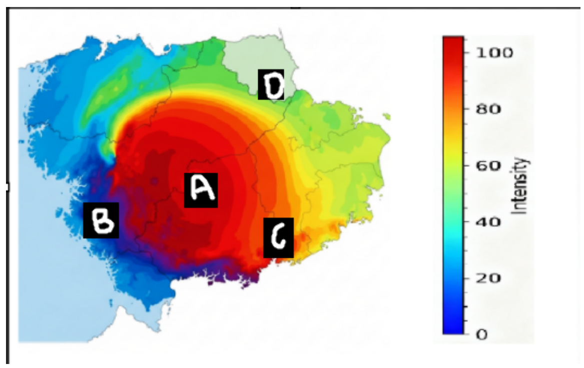
In this harsh world, even a shelter must be built with care and calculation. The strong winds outside create large pressure differences, which cause normal stress on the walls. This stress is given by the formula **σ = F/A**, where ***σ* is the normal stress**, ***F* is the applied force**, and ***A* is the area** over which the force acts. To make the walls stable, survivors use triangular metal frames based on the geometric relation **2r + c = a + b**, where *r* is the inradius and a,b,c are the lengths of the three sides (c is the hypotenuse). This ensures that the load is evenly distributed across all sides and also because the triangle is the ***strongest*** shape in nature (reference to Boss Baby).

But things are much worse outside the shelter. The air outside is always moving through convection - ***hot air rises, and cold air moves toward it,*** creating fast, unpredictable winds. Even sound behaves differently in such conditions. The **speed of sound is directly proportional to the square root of temperature and inversely proportional to the square root of density**. This means that when the air gets hotter, sound travels faster; when it becomes denser, sound slows down.

In this world, understanding these relationships isn't just science, it's survival. The right calculations can mean the difference between a shelter that stands firm and one that collapses under pressure. But lucky for you, you are a smart 9th grader…so let's beat this challenge for survival.

Space for Rough Work

Q1. The conditions are ever changing. You would need to transport a lot. What's better than your house being able to fly itself! Just pick the right spot to set up your shelter so that travel is the easiest.

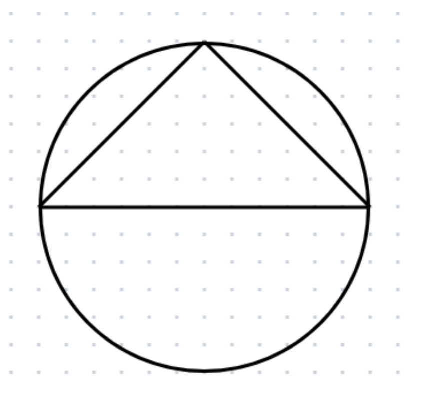


A) A
B) B
C) C
D) D

Space for Rough Work

Q2. A triangular shelter is built within a circular protective layering that has a diameter of 32m. One wall of the shelter is built along the full diameter of the circle. The total area of the shelter's floor plan is 256 m². An emergency operations room must be installed at a location that is equidistant from all three shelter walls. What is the distance of the room from each wall?

$$\frac{32^2}{2} \approx 22^2$$

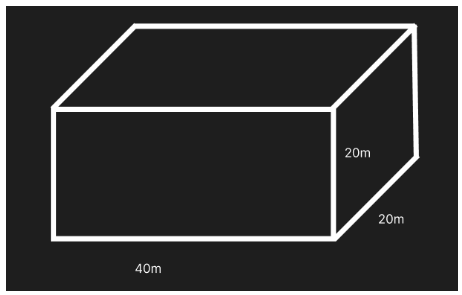


A) 5
B) 6
C) 7
D) 8

Space for Rough Work

Q3. The shelter wall receives a constant wind force of 50,000 N. Its composite material is rated for up to 150 N/m² normal stress. The shelter can be rotated so the wind hits either the 40m x 20m wall or one of the 20m x 20m walls. Sudden gusts may increase the force to 120,000 N.

Which side is safer as of now and what should be the normal stress so that the weaker wall does not break due to the sudden gust?



A) The 40m x 20m wall is safer; the material should withstand at least 300 N/m².

B) The 20m x 20m wall is safer; the material should withstand at least 150 N/m².

C) The 40m x 20m wall is safer; the material should withstand at least 150 N/m².

D) The 20m x 20m wall is safer; the material should withstand at least 300 N/m².

Space for Rough Work

Q4. The flight distance of the entire shelter depends on the following formula. For every 10% increase in the wind speed, 20% increase in time, 5% increase in altitude, find out the change in the flight distance. (10% increase in altitude decreases air density by 40% and increases the lift by 20%) (use 1.05/0.8 = 1.3)

$$D \propto \frac{v^2 \cdot t \cdot L}{\rho}$$

v is wind speed

t is time

L is lift variable

ρ is density

A) 89% increase

Space for Rough Work

B) 98% increase
C) 89% decrease
D) 98% decrease

Q5. In this world without agreed-upon maps or coordinates, you rely on sound to estimate how far away essential supplies are. You receive a signal: the sender tells you how long it should take for their shout to reach you—*but* this time is calculated using the speed of sound in Earth's atmosphere.

The catch? The air around you is 0.25 times denser than Earth's air and the temperature is 4 times than Earth's.

Suppose the shared signal travel time is t seconds.

What is the radius of the circle on which your supplies could be?

A) 4*π*v*t
B) 4*v*t
C) v*t
D) v/16 * t

Q6. Now that you have found out the distance of its location, you need to find its angle to pin point his exact location. (THIS IS HOW POLAR COORDINATES WORK BY THE WAY!). Somehow you get to know that your supplies have been moving at uniform speed on that circle for 45 days. You know that he completes one lap in 18 days. Consider they started directly from your north-east (east is 0 degree). Where are they now (in degrees)?

Space for Rough Work

A) 45
B) 0
C) 225
D) 315

Space for Rough Work

I MISS MY PET FISH

The Earth you know is gone. Ten millennia ago, the old world burned and drowned, a slow and agonizing end brought on by the carelessness of its own inhabitants. Now, the Earth is a canvas of scars and strange beauty. The seas have swallowed the continents, turning old nations into forgotten archipelagos where mutated corals grow in the flooded ruins of skyscrapers. The atmosphere, once a delicate balance, is now a swirling chaos of extreme weather. Tornadoes of fire rip across sun-scorched plains, while colossal hurricanes, known as "Leviathans," churn the new oceans for months on end. The forests are no longer made of docile pines but of towering, carnivorous flora with shimmering, bioluminescent leaves.The abundance of adverse conditions has created nightmarish creatures,straight from the depths of hell. It is in this harsh landscape that you,one of the last remaining humans,once lords and masters of this world, have to survive.

Humans were once at the peak of the biological ladder. That holds true no more.The harsh ever changing conditions have given rise to hellish creatures much better adapted to surviving the merciless climes.

In the sun-blasted plains and shattered deserts, one of the few signs of life that does not cower in the shade is the mirror backed grazer. Resembling an armadillo, its body is protected by a segmented carapace of polished, crystalline plates. This living shield, a marvel of post-apocalyptic evolution, acts as a perfect mirror, reflecting the unfiltered, searing radiation from the sky. As they slowly trudge across the scorched earth, grazing on the toxic, leathery flora, their backs send blinding flares of light into the sky.When threatened,the mirror backed grazers align their shells in such a way that they create multiple images of each animal.This makes the herd look more numerous than they actually are and scares off potential predators.

Space for Rough Work

Far from the poisoned seas and the sun-scorched plains, where the fractured remnants of mountains pierce the toxic sky, lurk the Kyrene. These aren't creatures of flesh and bone, but of living, solid crystal, their forms refracting the dim, filtered light into a terrifying mosaic of colors. They are arachnids in form, with eight long, spindly legs that click and scrape against the jagged rock. They communicate not with sounds we can understand, but with a series of high-pitched electrical pops and intricate whistles. These beings are the masters of piezoelectricity, generating powerful electrical charges through the very movement of their crystal bodies, a silent, deadly language that is beyond human comprehension. The Kyrene are the perfect, crystalline hunters of the new world, a lethal nightmare made real.

In the desolate, rust-colored wastelands, deadly creatures known as Auroras drift on the toxic winds. These ethereal hunters are not solid beings, but shimmering, translucent wraiths of pure energy that pulse with the faint, otherworldly glow of their namesake. Their essence is a powerful, localized magnetic field that they can intensify at will. A lone Aurora is a formidable predator, capable of disabling the primitive electronics of a scavenger's vehicle with a single, silent pass. But when they gather, their combined power is a terrifying spectacle. Gangs of Auroras move together in a silent coordinated hunt, and the potent magnetic field they generate is strong enough to rip ferrous metal from the ground. As they sweep across the landscape, they leave behind a destructive wake, raising a corrosive maelstrom of rust

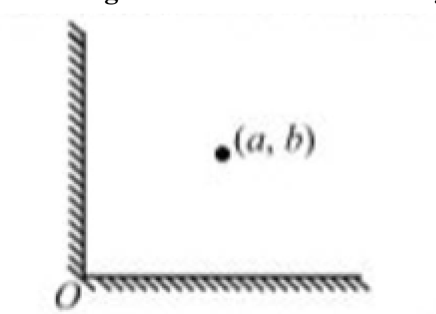
The Kyrene work collaboratively to send messages across large distances to other Kyrene.A single Kyrene produces a detailed message in the form of a high frequency EM wave,successive Kyrene step down the frequency by half each.Eventually,a sufficiently low frequency wave is produced which is then propagated in the necessary direction.

Space for Rough Work

Q1. Given that the initial Kyrene produces a wave of frequency 65.536 KHz,and a further 8 Kyrene are passed the message before it is propagated,how long would this message take to reach its destination 49*10^4 Km away(Wavelength=2.734 m).Answer to the closest integer.

A)700 S
B)7000 S
C) 70 S
D) 70000 S

Q2. If two mirror backed grazers have aligned their (assume plane mirror) shells at an angle of 90 degrees,and a third mirror backed grazer is standing exactly at the bisector of the angle,to a predator ,how many of these animals will they see(excluding the two that are reflecting the images)



A)2
B)6

Space for Rough Work

C)4
D)3

Q3. Auroras exert strong magnetic fields around themselves. It is necessary to wear strong protection when passing through areas of Aurora migration.With which materials should you not construct your protection?

A)Carbon Fiber
B)Copper
C)Lead
D)Aluminium

Q4. The Kyrene love lightning. They derive huge bursts of energy from being struck by lightning.The Kyrene also possesses the unique ability to vary their internal resistivity.During a lightning storm,the Kyrene with which combination of physical and electrical characteristics will receive the most lightning?

A)Low resistivity,Stout Body
B)High Resistivity, Stout Body
C)Low Resistivity,Thin Body
D)Low Resistivity,Thin Body

Space for Rough Work

Q5. The Kyrene reproduce via a suicidal process known as a mind meld.A large group of older Kyrene come together in an explosive process where the initial Kyrene are destroyed and each Kyrene pairs with every other Kyrene to produce one offspring . In one particularly large mind meld,639 adult Kyrene came together,how many offspring are produced?

A)203841
B)208341
C)204831
D)203481

Q6. Auroras are an invasive species that quickly take over any environment that they are introduced to.This is due to their nature of reproducing at death.Every Aurora splits into S new auroras at the time of natural death. This causes their population to multiply very rapidly .If initially there are n aurora and an aurora dies a natural death every x seconds, how many auroras will exist after t seconds?

A)$nS^{t/x}$
B)$tS^{x/n}$
C)$tSnx$
D)$st^{n/x}$

Space for Rough Work

HOPEFULLY NOT OCEANGATE

In this fractured world, movement and signal are luxuries few can afford. The survivors rely on the ***Cthonic Dredges*** - enormous pressure-resistant transports that crawl across the shifting plains and sunken highways of the drowned continents. Each dredge must withstand massive pressure variations as it ascends and descends the uneven terrain, where pressure is determined by ***P=ρgh*** where h is how deep the dredge is, ***ρ*** is the density and g is the acceleration due to gravity.

The onboard stabilizers monitor pressure as a function of height, tracing a linear relationship between the two variables.

(just as y(x)=a+bx describes any uniform change) ***[y(x) means y as a function of x]***

Over here, ***a = intercept*** and ***b = slope***. Slope is a measure of the quantity by which y would change for a unit change in x.

If we have two points (x1,y1) and (x2,y2), this same relationship can be presented in this way

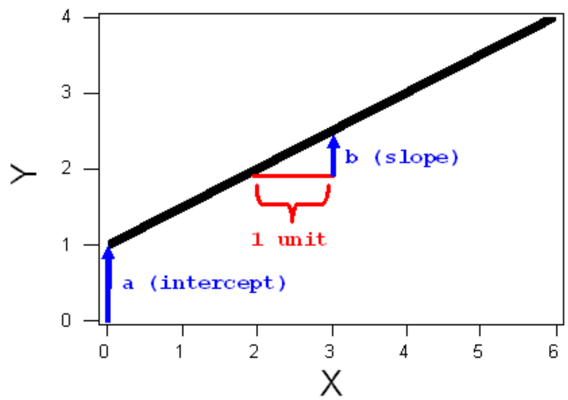
$$y(x) = y_1 + \frac{(y_2 - y_1)}{(x_2 - x_1)}(x - x_1)$$

Eg. If a quantity m, changes from m1 to m2 from t = 0 to t = T. Its change as a function of t can be written as

Space for Rough Work

$$m(t) = m_1 + \frac{(m_2 - m_1)}{(T - 0)}(t - 0)$$

[as point 1 is (m1, 0) and point 2 is (m2, T)]



But the storms do not only crush, they disrupt communication. In the absence of satellites, survivors encode their transmissions using ancient number systems, relics of human logic. Fundamentals of number systems come into play.

Space for Rough Work

Number systems work on a very fundamental system of bases. The decimal system has a base of 10, binary system has a base of 2, octal system has a base of 8, etc.

Number = sum of place values of each digit

$$\text{Place value of a digit} = (\text{digit}) \times (\text{base})^{\text{index}}$$

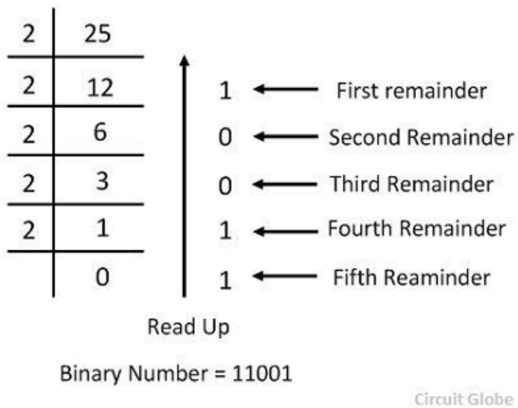
[index starts from the right most side from 0]

For eg. 3710 = 7*10^0 + 3*10^1

10012 = 1*2^0 + 0*2^1 + 0*2^2 + 1*2^3 = 1 + 0 + 0 + 8 = 9

To convert from decimal to any system, simply keep divide by the base and read the remainders up. Refer to this image to figure out conversion of decimal to binary.

Space for Rough Work



Q1. A transport wheel is a hollow cylinder of radius 20cm. It rolls without slipping. Each full rotation compresses a piezoelectric strip generating 15 J of electricity. If the survivor travels 40π m. How many rotations does the wheel make and how much total energy is generated?

a. 80 rotations, 1200J

Space for Rough Work

b. 10 rotations, 150 J
c. 1000 rotations, 15000 J
d. 100 rotations, 1500 J

Q2. In this post-apocalyptic world one also has to build a submarine that allows us to travel underwater and avoid the leviathans. Apocalypse has led to variation in the density of the seawater across the journey we plan. It varies from ρ_1 to ρ_2 uniformly with respect to time as we travel across the seas (Densities are in the units kg/m^3). The duration of the journey will be in T hours

To accommodate for this change in density the submarine must intake mass (mass intake rate=μ_m(kg/hour)). Let the volume of the submarine be $V(m^3)$ and acceleration due to gravity be $g(m/sec^2)$. Find μ_m.

A) $\mu_m=V(\rho_2-\rho_1)/T$

B) $\mu_m=V(\rho_2-\rho_1)g/T$

C) $\mu_m= V(\rho_1-\rho_2)/(2T)$

D) $\mu_m=V(\rho_1-\rho_2)g/(2T)$

Space for Rough Work

Q3. The submarine must also be capable of surviving under high pressure. The machine also must avoid turbulence underwater. Therefore, it has mapped its route across the ocean. The submarine will start its journey at a depth d_1 and end its journey at a depth d_2(ascending uniformly with respect to time over the journey). Calculate the pressure(P) at any time t experienced by a certain point on the machine.

A) $\left[d_2t - d_1(t-T)\right]\left[\rho_2 t - \rho_1(t-T)\right] g$

B) $P = \dfrac{\left[d_2t - d_1(T-t)\right]\left[\rho_2 t - \rho_1(t-T)\right] g}{T^2}$

C) $P = \dfrac{\left[d_2t - d_1(t-T)\right]\left[\rho_2 t - \rho_1(T-t)\right] g}{t^2}$

D) $P = \dfrac{\left[d_2t - d_1(T-t)\right]\left[\rho_2 T - \rho_1(T-t)\right] g}{T^2}$

Q4. Which description best fits the motion of the submarine in this graph?

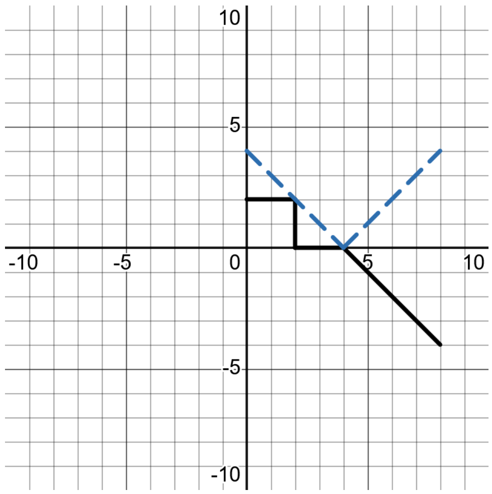
y-axis : magnitude

x-axis : time

Blue-dotted : pressure

Space for Rough Work

Black-solid : velocity



A) The submarine ascends, comes to a stop, then descends at a uniform acceleration.

B)The submarine descends, comes to a stop, then ascends at a uniform acceleration.

C) The submarine ascends then descends almost instantly.

D) The submarine descends and then ascends with increasing speed.

Q5. A secret message was encoded using a ternary system (base-3), but the decoding team mistakenly read it as binary (base-2). If the code decoded by 28. What was the actual number they wanted to encode?

A) 117
B) 13
C) 28
D) 119

Q6. The transport device's ternary control system is upgraded so that each digit can now represent directional thrust:

- **0 = "no thrust"**
- **1 = "thrust forward (+1)"**
- **2 = "thrust backward (–1)"**

Each 3-digit ternary code determines the net thrust (sum of all digit values).

If the device must generate a net thrust of exactly +1, how many valid ternary codes exist?

A) 6

B) 7
C) 8
D) 9

Space for Rough Work

WHY IS THE BEACH CLOSED..?

The Earth you know is gone. Ten millennia ago, the old world burned and drowned, a slow and agonizing end brought on by the carelessness of its own inhabitants. Now, the Earth is a canvas of scars and strange beauty. The seas have swallowed the continents, turning old nations into forgotten archipelagos where mutated corals grow in the flooded ruins of skyscrapers. The atmosphere, once a delicate balance, is now a swirling chaos of extreme weather. Tornadoes of fire rip across sun-scorched plains, while colossal hurricanes, known as "Leviathans," churn the new oceans for months on end. The forests are no longer made of docile pines but of towering, carnivorous flora with shimmering, bioluminescent leaves.The abundance of adverse conditions has created nightmarish creatures,straight from the depths of hell. It is in this harsh landscape that you,one of the last remaining humans,once lords and masters of this world, have to survive.

The oceans, once a source of life and wonder, have become a boiling, venomous cauldron. Their surface is a perpetual, greasy film of discarded oils and ancient chemicals that have risen from the seabed. Beneath this film, the waters are a patchwork of deadly thermal climes, from freezing depths kept alive by geothermal vents to vast, scalding currents heated by underwater volcanic eruptions. This constant thermal chaos breeds colossal, unpredictable tsunamis that can appear from nowhere, slamming into the fractured coastlines and washing away entire settlements.

Space for Rough Work

The raging seas are warmed by underwater volcanic activity which supplies them with immeasurable amounts of heat every second.However ,water rises to the surface and cools off ,allowing the seas to remain in a precarious balance.

Q1. Which of the following processes drives this cooling of the seas?

a)Evaporation
b)Convection
c)Condensation
‘ d)Both a) and b)

Q2. Huge whirlpools dominate the surface of the seas, looking like giant maws of the ocean, ready to grab and devour any unsuspecting beings unfortunate enough to stray into their grasp. On one occasion ,your submarine accidentally gets stuck in the outermost circle of one of the largest whirlpools on the planet. You continue to spin round and round for days on end. Your engineer claims he can provide a sudden burst of force in exactly one direction to try to free you from the whirlpool.In which direction should your engineer apply the force?

a)Radially Outwards
b)Radially Inward
c)Along your direction of motion in the circle(tangentially forward)
b)Against your direction of motion (tangentially backwards)

Space for Rough Work

Q3. If accidentally you moved into an inner ring of the whirlpool and were stuck there,going around in circles at the same speed as the outermost circle , compare the force needed to travel from this circle to the next circle with greater radius and the force needed to escape the whirlpool from the greatest radius circle.

a)Inner Force<Outer Force
b)Outer Force<Inner Force
c) Outer Force=Inner Force
d)Insufficient Information to decide

Q4. While returning from a deep sea exploration sojourn, you observe a naturally formed bubble in the sea from your submarine's window. As you rise to the surface at the same rate as the bubble, you see that the bubble grows in size. Explain why this happens?

a)The air inside the bubble gets hotter
b)The air inside the bubble gets cooler
c)External Water pressure decreases at the surface
d) The air inside the bubble changes in chemical composition

Space for Rough Work

Q5. Since air is all around you as well in the submarine,should you or should you not be afraid of this air showing a similar phenomenon as the air bubble? Why or why not?

a)Worried,the submarine will expand like the bubble and explode
b)Not worried,metal walls of the submarine insulate it from everything external
c)Worried,the chemical composition of air will change and humans won't be able to breathe it
d)Not worried, humans constantly breathing does not allow the chemical composition to change.

Q6. Your submarine has three layers as part of its walls to protect you from the hazards of the ocean's heat.
i)A layer of thick wool like substance in between metallic walls
ii)An vacuum gap between two metallic walls
iii)A silvered coating on the outer side of the outermost metallic walls

What methods of heat transfer are these safeguards placed to prevent respectively?
a)Conduction,Convection,Radiation
b)Convection,Radiation,Conduction
c)Radiation,Convection,Conduction
d)Conduction,Radiation,Convection

Space for Rough Work

Q7. Enough time has elapsed since the creation of the earth from cosmic gas clouds at the formation of the solar system. Assume it abruptly stops spinning. Choose the most appropriate consequence.

a)Life continues as before, no effect.
b)Earth flies away from the sun,out of the solar system
c)Massive tsunamis arise from the oceans travelling eastwards,wrecking everything in their path.
d)Gravity crushes everything to earth's surface as there is no longer any outward force.

Space for Rough Work

ANSWER KEY

Resources Low, Morale Lower													
1	A	2	B	3	A	4	A	5	B	6	B		
I miss my Pet Fish													
1	A	2	C	3	D	4	A	5	A	6	A		
Hopefully not Oceangate													
1	D	2	A	3	D	4	A	5	A	6	A		
Why is the Beach closed?													
1	D	2	A	3	B	4	C	5	B	6	A	7	C

SOLUTIONS

RESOURCES LOW MORALE LOWER

1. *Correct answer - A)*

   A *is correct because it is the hottest, highest-intensity area. Strong heating creates the greatest pressure differences, which drive* ***stronger wind movement****. Those stronger winds make it easiest for the flying house to lift and travel efficiently.*

2. *Correct Answer -B) 6*

   Start by using the given floor area of the triangular shelter to relate the two legs of the triangle, which determines the product ***a*b****. Since the triangle is built along the full diameter of the circular layer, it must be a right triangle, so the side lengths satisfy* $a^2+b^2=c^2$

   Combining this with the known value of ***a*b****, evaluate* $(a-b)^2=a^2-2a*b+b^2$ *(which will give a=b upon calculation)*

*With this symmetry established, applying the given geometric relation **2r+c=a+b** gives us the value of r.*

3. *Correct Answer - A) The 40m x 20m wall is safer; the material should withstand at least 300 N/m².*

   To find the safer side, apply the principle that stress is inversely proportional to area; since the 40m×20m wall provides a larger surface to distribute the constant force, it results in a lower stress concentration than the smaller wall.

   To ensure the weaker wall survives a gust, calculate the stress by dividing the maximum gust force by the smaller wall's area. This value represents the minimum rating the material must have to prevent failure on the side with the highest pressure intensity.

4. *Correct Answer - A) 89% increase*

   By substituting new values into the formula, the combined product of the speed, time, and lift multipliers is divided by the reduced density multiplier. The resulting value represents the new distance as a ratio of the original; subtracting the baseline reveals the total percentage increase in flight distance.

5. *Correct Answer - B)* 4*v*t

   To determine the radius of the circle where supplies could be located, we must find the actual speed of sound in the current environment compared to Earth's speed of sound (v) and then find the total distance (which is speed times time taken)

   Using the relationship given in the intro paragraph we can see that the speed of sound in your current environment is exactly four times faster than v. Since the radius of the circle is simply the product of the actual speed and the travel time (t), the supplies are located at a distance of $4 \cdot v \cdot t$.

6. *Correct Answer - B) 225*

   To find the final position, determine the initial angle and the total angular change over the travel period. Since the journey starts at north-east and east is 0°*, the starting position is* 45°*. Next, calculate the number of laps by dividing the total travel time by the period of one revolution, which results in two full rotations plus an additional half-lap. Since a half-lap equals* 180°*, adding this displacement to the* 45° *starting point places the supplies at a final coordinate of* 225°.

I MISS MY PET FISH

1. *Correct Answer - A) 700S*

Using the given wavelength and frequency gives a propagation speed on the order of 10510^5105 m/s. Over an interplanetary-scale distance of 10810^8108 meters, this leads to a travel time closest to a thousand seconds.

2. *Correct Answer - C) 4*

   Two plane mirrors at 90° produce 3 images. Including the real third grazer, the predator sees 4 animals.

3. *Correct Answer - D) Aluminium*

   Strong magnetic fields induce currents in good conductors. Aluminium is highly conductive and unsafe here.

4. *Correct Answer - A) Low resistivity, Stout Body*

   Lightning favors low resistivity paths and larger targets. A stout body with low resistivity attracts the most strikes.

5. *Correct Answer - A)* 203841

   Each Kyrene produces one offspring with every other Kyrene exactly once. So the first Kyrene pairs with 638 others, the next with 637 new ones, and so on, until the last has none left to pair with. Adding all these pairings gives the total number of offspring.

6. *Correct Answer - A)* $nS^{t/x}$

   Each time an Aurora dies, it is replaced by ***S new ones****, so the population multiplies rather than just increasing. Repeating this every x seconds causes exponential growth over time.*

HOPEFULLY NOT OCEANGATE

1. *Correct Answer - D)100 rotations, 1500 J*

   To determine the total energy generated by the transport wheel, we first find the number of rotations by comparing the total travel distance to the wheel's circumference. Since the wheel rolls without slipping, each rotation covers a linear distance equal to 2π times its radius. After converting the radius to meters, dividing the total travel distance by this circumference reveals that the wheel completes exactly 100 rotations.

The total electricity produced is then found by multiplying the number of full rotations by the energy generated per compression. With each rotation contributing 15 J of electricity, the 100 rotations result in a total energy output of 1500 J.

2. *Correct Answer - A)* $\mu_m = V(\rho_2 - \rho_1)/T$

   Since the density changes uniformly from ρ1 to ρ2 over a total time T, we first find the total change in mass required by calculating the difference between the final displaced mass (V·ρ2) and the initial displaced mass (V·ρ1). Dividing this total mass change by the duration of the journey (T) gives the rate at which mass must be taken in. Factoring out the volume leads to the final expression

   Because the submarine must stay submerged, its total mass must always equal the mass of the water it displaces, which is the product of its constant volume (V) and the current water density.

3. *Correct Answer - D)*$P=[d_2t-d_1(T-t)]\,[\rho_2T-\rho_1\,(T-t)]g/T^2$

   Since the submarine's depth and the seawater's density both change uniformly with respect to time, we express each as a linear function. By substituting these time-dependent equations into the pressure formula, we arrive at the final algebraic expression.

$$d(t) = \frac{d_1(T-t) + d_2 t}{T}$$

$$\rho(t) = \frac{\rho_1(T-t) + \rho_2 t}{T}$$

4. *Correct Answer - A) The submarine ascends, comes to a stop, then descends at a uniform acceleration.*

   *Initially, the velocity is constant and positive, causing the pressure to decrease linearly; this indicates the submarine is **ascending** toward the surface. Following this, the velocity drops to zero between t=3 and t=5, meaning the submarine **comes to a stop** at its minimum depth.*

   *Finally, the velocity increases linearly in the negative direction, representing **uniform acceleration** as the submarine moves downward. This is confirmed by the rising pressure line, which shows the submarine is **descending** back into the depths.*

5. *Correct Answer - A) 117*

   *To decode the message, first convert the decimal number **28** back into the binary digits the team mistakenly read. Since 28 is the sum of 16,8, and 4, the binary sequence is **11100**.*

Next, re-evaluate this same sequence using the correct ***ternary (base-3)*** *system. By calculating the powers of three ($1\times81+1\times27+1\times9+0+0$), the actual encoded number is revealed to be* ***117****.*

6. *Correct Answer - A)* 6

   Since the net thrust is the sum of all three digit values, there are two distinct ways to achieve a total of +1:

   - ***Case 1: One forward thrust and two "no thrusts"*** *The values used are {1,0,0}. Because these can appear in any order within the 3-digit code, there are* ***3 permutations****: (1,0,0), (0,1,0), and (0,0,1).*
   - ***Case 2: Two forward thrusts and one backward thrust*** *The values used are {1,1,−1}, which sum to +1. Similarly, these can be arranged in any order, resulting in* ***3 permutations****: (1,1,2), (1,2,1), and (2,1,1). (Note: in ternary code, the value −1 is represented by the digit* ***2****).*

WHY IS THE BEACH CLOSED..?

1. *Correct Answer - D) Both a and b*
   Hot water rises and cooler water sinks, setting up convection currents, while evaporation at the surface removes heat as high-energy molecules escape. Both processes together help balance the ocean's heat.

2. *Correct Answer -A) Radially Outwards*
   To escape circular motion, you must overcome the inward pull keeping you trapped. A force applied directly outward works against this and lets you spiral away.

3. *Correct Answer - B) Outer Force<Inner Force*
   At a smaller radius, the inward force needed to maintain circular motion is larger for the same speed. Escaping from the inner ring therefore requires more force.

4. *Correct Answer - C) External Water pressure decreases at the surface*
   As the bubble rises, the surrounding water pressure decreases. With less external pressure squeezing it, the air inside expands and the bubble grows.

5. *Correct Answer - B) Not worried,metal walls of the submarine insulate it from everything external*
   The air inside the submarine is not free to expand like a bubble because it is enclosed by rigid walls. Its volume stays fixed, so no dangerous expansion occurs.

6. *Correct Answer - A) Conduction,Convection,Radiation*

Wool traps air and reduces conduction, a vacuum stops both conduction and convection, and silvered surfaces reflect thermal radiation.

7. *Correct Answer - C) Massive tsunamis arise from the oceans travelling eastwards,wrecking everything in their path.*
   The oceans retain their eastward motion due to inertia, causing massive global tsunamis that sweep across continents.